

Lab 22 Worksheet — Applying the CompTIA Six-Step Troubleshooting Methodology

Name: ____ Date: ____

Exam objective: Apply the CompTIA six-step troubleshooting methodology in order to a real fault, documenting each step (Core 1 objective 5.1).

Record as you go

Step	What you did	What you observed
1	Write the six steps in order before you start, because the order itself is examinable: identify the problem; establish a theory of probable cause; test the theory; establish a plan of action and implement it; verify full functionality and implement preventive measures; document findings, actions and outcomes.	
2	Step 1 — identify the problem. Interview the user with open questions: what exactly happens, when did it start, what changed, and can you reproduce it on demand.	
3	Step 1 continued — always ask what changed, because most faults follow a recent change: a new driver, an update, a moved cable or new software.	
4	Step 1 continued — back up user data before making any change, so that your troubleshooting can never be the cause of data loss.	
5	Step 2 — establish a theory of probable cause, questioning the obvious first and listing at least three candidate causes ranked by likelihood.	
6	Step 3 — test the theory. Design a test that will disprove the theory if it is wrong, not merely one that confirms what you already believe.	

Step	What you did	What you observed
7	Step 3 continued — if the theory is not confirmed, establish a new theory or escalate. Record that escalation is a legitimate step in the methodology, not a failure.	
8	Step 4 — establish a plan of action, referring to vendor documentation, and state the rollback position before you implement anything.	
9	Step 5 — verify full system functionality, testing not only the reported fault but the functions around it that your change might have affected.	
10	Step 5 continued — implement preventive measures so the same fault does not recur, and state the specific measure you applied.	
11	Step 6 — document findings, actions and outcomes in the ticket, in language a colleague could follow without speaking to you.	
12	Practise the discipline on the Killercoda playground by diagnosing a deliberately broken DNS configuration through all six steps and recording your evidence.	

Verification

Success criterion: Your record contains all six steps in the correct order, each with written evidence; a rollback position is stated before implementation; a specific preventive measure is named; and the DNS exercise shows the fault reproduced and then resolved.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
